

Recursive Offline Distillation via Epistemic Complexity Approximation

Anonymous ACL submission

Abstract

Distillation of reasoning chains from Large Language Models (LLMs) into their smaller siblings is a standard training approach. Frequently researchers do not have full access to a teacher model and its internal representations, which limits the resulting performance. We propose a novel blueprint for efficient offline distillation that recursively uses lightweight epistemic and aleatoric complexity approximations for selective distillation. Specifically, for three open models (3-4B) at each epoch we estimate epistemic uncertainty using a medium model (70-72B) as an aleatoric proxy and sample only a small subset of the reasoning chains for distillation. Our approach consistently outperforms naive distillation on comparable computational budget (57.15 ± 1.01 vs 51.61 on average) and offers a much steeper learning curve. Additionally, we present a comparative analysis of methods to generate and normalize distillation traces. We publish our code and data¹.

1 Introduction

Modern LLMs come in different sizes, and their performance is often correlated with the number of parameters. The largest of them demonstrate exceptional results and span 1-2 trillion parameters. Further domain adaptation (Cheng et al., 2024; Lu et al., 2024) or even inference at this scale requires massive computational resources (Wan et al., 2024) unavailable to most practitioners. It prompts the need for capable smaller models (Chen and Varoquaux, 2026) that can be fine-tuned and deployed in resource-constrained environments.

A prominent approach for small model training is distillation from a larger teacher model (Hinton et al., 2015; Sanh et al., 2020; Xu et al., 2024). Many methods rely on having access to the online teacher (Agarwal et al., 2024; Jiao et al., 2020; Gu et al., 2024), which is not always possible (Zhou

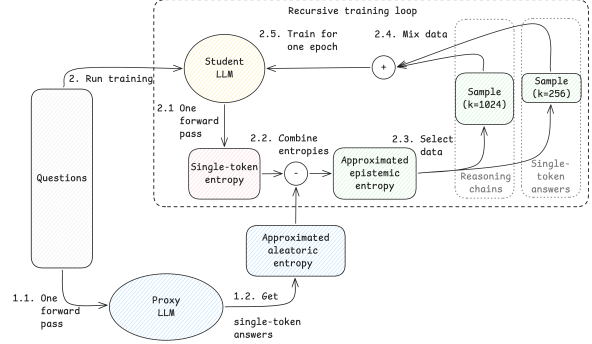


Figure 1: Recursive offline distillation scheme

et al., 2023; Ryskulov et al., 2026). It forces practitioners to explore offline distillation, where the student model is trained on the pre-generated teacher’s text output (Li et al., 2023; Hsieh et al., 2023).

We propose a fully automated recursive pipeline for offline distillation (Figure 1, described in 3) based on uncertainty decomposition (Wang et al., 2025b). The method consistently outperforms naive distillation (57.15 ± 1.01 vs 51.61 on average), and it is designed to work in resource-constrained environments.

Additionally, we provide a comparative analysis of augmentation strategies for synthetic reasoning chains when the teacher model does not perform perfectly on the target distillation domain and conduct sensitivity studies of the synthetic trace generation and normalization methods and sampling strategies.

2 Related Work

Recent work improved distillation through training objectives (Ko et al., 2024; Xu et al., 2026), data selection (Just et al., 2026; Li et al., 2025; Yang et al., 2026), and curriculum learning (Jiang et al., 2025; Zhang and Liu, 2026). Joint supervision with rationales and labels improves data efficiency (Hsieh et al., 2023). Curriculum benefits vary with model capability, task complexity, and the choice

¹TODO

of difficulty metric (Jia et al., 2026). ELAD (Zhang et al., 2024) selects uncertain student explanations for teacher revision. Complexity-aware fine-tuning (Goncharov et al., 2026) selects reasoning examples by single-token entropy. We recursively sample pre-generated traces using student-proxy entropy differences.

Trace synthesis includes explaining known solutions (Li and Mooney, 2024; Wang et al., 2025a) and learning from corrections (An et al., 2024; Yang et al., 2025). We compare direct generation and various augmentations on the same task under constrained trace lengths.

3 Method

Our central hypothesis is that we can approximate epistemic uncertainty of the student model with its entropy and entropy of a larger LLM. Goncharov et al. (2026) show that a single-token entropy (Figure 4) is a reasonable complexity predictor for a given entry in a question-answering dataset. We assume that it approximates the total uncertainty and that we can decompose it into epistemic and aleatoric components (Depeweg et al., 2018; Malinin and Gales, 2021). With scaling laws (Kaplan et al., 2020) in mind, we hypothesize that larger models should have lower total uncertainty. Therefore, in comparison to a small student model (3-4B) its substantially larger sibling (70-72B²) could be treated as a weak aleatoric proxy.

For a question x , assume $H(x) = A(x) + E(x)$, where H_S , H_P - single-token entropy (student and proxy), A - shared aleatoric component, and E_S , E_P - model-specific epistemic component. If proxy’s residual epistemic uncertainty satisfies $E_P(x) \leq \varepsilon(x)$, then

$$\begin{aligned} A(x) &= H_P(x) - E_P(x) \geq H_P(x) - \varepsilon(x) \\ E_S(x) &= H_S(x) - A(x) \leq H_S(x) - H_P(x) + \varepsilon(x) \end{aligned} \quad (1)$$

Thus, the entropy difference (*entropy gain*) approximates the student’s epistemic uncertainty with error at most $\varepsilon(x)$.

The proposed pipeline consists of the following steps (Figure 1, Algorithm 1): (1) prompt the larger model to answer all questions in the dataset with a single token once (or an ensemble of models taking average), collect entropy, assume it is an aleatoric uncertainty proxy; (2) for each training

Algorithm 1 Recursive offline distillation via epistemic complexity approximation

Require: Student S , proxy model P , epochs T
Require: Questions D , ground-truth answers y_x , and pre-computed reasoning traces r_x
Ensure: Trained student S

```

1: for all  $x \in D$  do
2:    $H_P(x) \leftarrow \text{ENTROPY}(P, x)$ 
3: end for
4: for  $t = 1, \dots, T$  do
5:   for all  $x \in D$  do
6:      $H_S(x) \leftarrow \text{ENTROPY}(S, x)$ 
7:      $\text{EG}(x) \leftarrow \max\{H_S(x) - H_P(x), 0\}$ 
8:   end for
9:    $Q \leftarrow \text{WEIGHTEDSAMPLE}(D, \text{EG}, 1024)$ 
10:   $Q_A \leftarrow \text{first 256 questions in } Q$ 
11:   $B_R \leftarrow [(x, r_x) : x \in Q]$ 
12:   $B_A \leftarrow [(x, y_x) : x \in Q_A]$ 
13:   $S \leftarrow \text{TRAIN}(S, B_R \parallel B_A)$ 
14: end for
15: return  $S$ 

```

epoch, (2.1) prompt the student model to answer all questions in the dataset with a single token, collect entropy; (2.2) calculate entropy gain $\text{EG}(x) = \max\{H_S(x) - H_P(x), 0\}$; (2.3.1) sample 1024 synthetic reasoning chains without replacement, weighted by positive entropy gain; (2.3.2) sample 256 single-token answers in similar fashion as a subset of the previous sample; (2.4) concatenate samples; (2.5) train the student using LoRA (Hu et al., 2021) on reasoning traces first, followed by single-token responses.

4 Results

4.1 Experimental Setup

Dataset: Multiple choice question answering dataset MMLU Pro (Wang et al., 2024). It has questions of different complexities from 14 domains, ten options for each, single correct one. Random train/test split - 80/20.

LLMs: Aleatoric proxy: ensemble of two medium models³ (taking average of their entropies) - LLaMA 3.3 70B (Grattafiori et al., 2024), Qwen 2.5 72B (Qwen-Team, 2024). Students: three small models Qwen2.5 3B (Qwen-Team, 2024), Phi-4-Mini (Microsoft et al., 2025), LLaMA 3.2 3B (Grattafiori et al., 2024). Reasoning trace source: DeepSeek V4 Pro and Flash (DeepSeek-AI et al., 2026) via API.

²This particular size is chosen based on the imposed resource constraints (80GB VRAM) and open model availability

³All models and datasets are published under permissive licenses that allow them to be used for research purposes

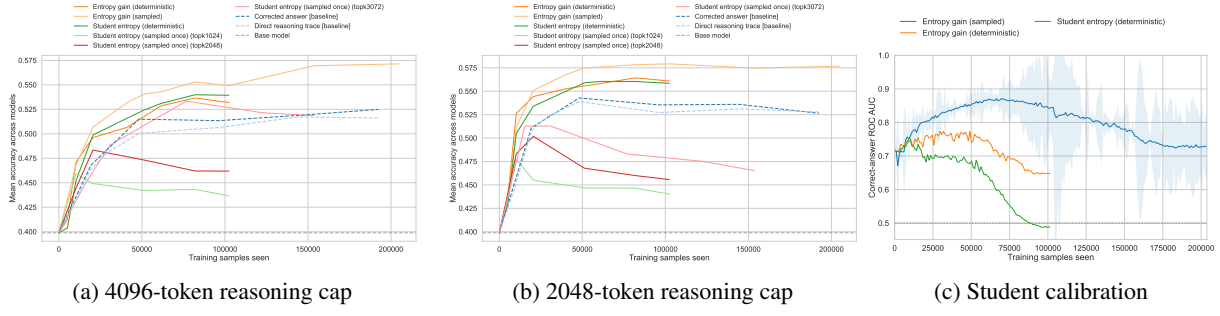


Figure 2: Distillation pipeline results across three models (average) on MMLU-Pro

Method	LLaMA 3.2 3B	Qwen 2.5 3B	Phi4-mini	Mean
Direct reasoning trace	46.09 / 45.55	51.45 / 50.37	57.27 / 62.43	51.61 / 52.78
Corrected answer	47.80 / 47.26	50.87 / 49.42	58.81 / 61.26	52.49 / 52.65
Student entropy (deterministic)	49.83 / 50.08	53.62 / 53.03	58.35 / 64.42	53.93 / 55.85
Entropy gain (deterministic)	<u>50.17 / 50.37</u>	51.04 / <u>53.28</u>	58.40 / 64.63	53.20 / <u>56.10</u>
Student entropy (sampled once, $k = 1024$)	41.35 / 40.11	39.44 / 39.07	50.17 / 52.87	43.65 / 44.01
Student entropy (sampled once, $k = 2048$)	42.19 / 41.06	43.18 / 43.02	53.20 / 52.62	46.19 / 45.57
Student entropy (sampled once, $k = 3072$)	42.02 / 42.81	56.77 / 40.86	56.77 / 55.90	51.86 / 46.52
Entropy gain (sampled, ours)	52.58 $\pm$ 1.94 / 52.08 $\pm$ 2.28	<u>56.64</u> $\pm$ 3.79 / 56.86 $\pm$ 2.32	62.22 $\pm$ 2.42 / 64.07 $\pm$ 4.52	57.15 $\pm$ 1.01 / 57.67 $\pm$ 1.19

Table 1: Distillation pipeline last epoch accuracy (4096-token / 2,048-token thinking budget) on MMLU-Pro.

Hardware: We run all experiments⁴ on a single A100/H100⁵ GPU with 80GB of VRAM to validate the method in resource-constrained environments.

4.1.1 Our method and results

We evaluate the proposed pipeline for 200 epochs (205k questions seen) and 2 thinking budgets (2048 and 4096 tokens). It is compared at similar data budget with:

Direct reasoning trace, Corrected answer - naive distillation on the most promising types of reasoning traces (see 4.3);

Student entropy (deterministic), Entropy gain (deterministic) - alternative pipeline with weighted sampling replaced with greedy sampling;

Student entropy (sampled once, $k=x$) - distillation executed against k samples selected once by student entropy (not recursively at each epoch).

Entropy gain (sampled), Student entropy (deterministic), Entropy gain (deterministic) use the proposed data mix (reasoning traces/single-token answers = 1024/256), other methods use only reasoning traces.

Notably, the proposed data mix increases the number of training tokens only by 0.003% - 256 extra tokens compared to 8,388,608 tokens in 1024 reasoning chains.

All randomly sampled methods use 3 random

seeds, reporting $\pm 95\%$ confidence-intervals, and deterministic methods are evaluated once with greedy sampling.

4.2 Main Results

Table 1 and Figure 2 show the results for MMLU Pro. We see that the proposed method produces a significant improvement over naive distillation (up to 10% gain). For both reasoning budgets, it provides the best result on average across three models in comparison with other approaches. Greedy sampling demonstrates subpar and less stable curves, though surpassing the baselines. Greedy distillation based on student entropy sampled once (Goncharov et al., 2026) shows signs of severe degradation after the initial uplift. However, Qwen 2.5 3B, notably, demonstrates the best performance with it at $k=3072$, but degrades at smaller sample sizes, revealing the fragility of the approach.

We also note that our method helps with the model calibration significantly enhancing ROC AUC of single-token entropy for the first half of training. Shortly after, it starts declining which is loosely correlated with the method’s plateau.

4.3 Synthetic Traces Generation

In resource-constrained environments, available VRAM imposes a strict limitation on the length of distilled reasoning chains. As soon we start truncating them, we face a new challenge - which part

⁴Training hyperparameters can be found in Appendix A.

⁵We had to switch cards depending on availability

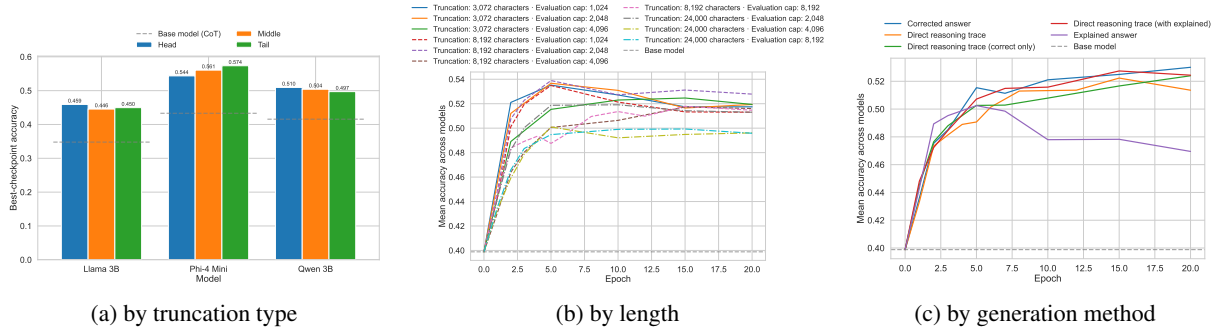


Figure 3: Comparison of synthetic traces across three models (average)

of the trace contains the most useful signal? At last, how exactly do we prompt the teacher model to generate reasoning traces?

We evaluate reasoning traces by: (a) *truncation type* - which part of the trace we keep when it exceeds the maximum length; (b) *length* - what is the optimal training truncation limit (characters) and evaluation thinking budget (tokens); (c) *generation method* (below).

Generation methods (details in Appendix C): (1) *Direct reasoning trace* - baseline prompt to just answer the question; (2) *Direct reasoning trace (correct only)* - same as (1), but we keep only traces with correct answers; (3) *Explained answer* - we give a correct answer to the model and prompt it to explain it; (4) *Direct reasoning trace (with explained)* - same as (1), but replace incorrect traces with (3); (5) *Corrected answer* - same as (1), but for incorrect traces we give the model correct answers and ask it to update its reasoning.

Figure 3 shows that models have a slight preference towards head truncation, medium to short reasoning chains in training and thinking budgets. Surprisingly, longer reasoning chains and greater thinking budgets lead to notable performance decrease. As to the generation methods, we see that augmented reasoning traces lead to elevated performance, while the consistent data leakage results in degradation. Based on the findings, we use 8,192-character head truncation and *Corrected answer* generation for the pipeline.

4.4 Sensitivity Study

To attribute performance gains to either the data mix or the entropy-based score and sampler, we compare the proposed pipeline (*entropy gain*) with: (1) alternative pipelines where we sample according to *student entropy* and *discounted entropy gain* (see below); (2) alternative pipeline where we use

Sampler	Three-model mean
Direct reasoning trace (baseline)	51.61
Random	55.08 ± 0.80
Student entropy	55.87 ± 1.40
Discounted entropy gain	<u>55.98</u>
Entropy gain	57.15 ± 1.01

Table 2: Distillation pipeline last epoch accuracy for different sampling methods (4096-token thinking budget)

a random sampler.

Discounted entropy gain is defined as $\max\{H_S(x) - k(t) * H_P(x), 0\}$, $k(t) \in [0, 1]$, where t - training epoch. We set $k(t)$ initially to 0.8 and decrease it according to a schedule, hypothesizing that as the student undergoes training the static medium model stops being a good aleatoric proxy.

Table 2 shows that across three models on average entropy gain provides the best results confirming the contribution of the aleatoric uncertainty approximation. At the same time, we notice that random sampler performs exceptionally well (confirming findings of Okanovic et al. (2023)) compared to the baseline, which points to the proposed data mix as a significant contributor to the gains.

5 Conclusion and Discussion

This paper introduces a recursive offline distillation method rooted in epistemic uncertainty approximation (57.15 ± 1.01 mean accuracy across three students on MMLU-Pro vs 51.61 for direct distillation, offers a much steeper learning curve (baseline score at 15% of its data budget)). We confirm that both the entropy difference sampler and the data mix contribute to the gains. Surprisingly, shorter traces and reasoning budgets perform better. Further exploration of the more rigorous epistemic uncertainty estimation and the surprising effect of the small data mix augmentation remain future work.

Limitations

- Method is tested only with small models. We do not know how well it is going to generalize to larger LLMs. Additionally, it might be challenging to find a suitable aleatoric proxy for larger students.
- We use a single dataset. We do not know how well it translates to other settings.
- Alternative aleatoric approximations, such as agreement of the ensemble and other, are left unexplored.
- We did not test and compare multiple sizes of aleatoric proxies and their impact on performance.
- Different compositions of the data mix are not tested due to lack of resources.
- We tested only a single schedule for *Discounted Entropy Gain*. Alternatives are yet to explore.
- Some low-resource environments might have even more constraints and therefore struggle to fit a suitable aleatoric proxy into VRAM.

References

Rishabh Agarwal, Nino Vieillard, Yongchao Zhou, Piotr Stanczyk, Sabela Ramos Garea, Matthieu Geist, and Olivier Bachem. 2024. [On-policy distillation of language models: Learning from self-generated mistakes](#). In *International Conference on Learning Representations*, volume 2024, pages 21246–21263.

Shengnan An, Zexiong Ma, Zeqi Lin, Nanning Zheng, Jian-Guang Lou, and Weizhu Chen. 2024. [Learning from mistakes makes llm better reasoner](#). *Preprint*, arXiv:2310.20689.

Lihu Chen and Gaël Varoquaux. 2026. [What is the role of small models in the llm era: A survey](#). *Preprint*, arXiv:2409.06857.

Daixuan Cheng, Shaohan Huang, and Furu Wei. 2024. [Adapting large language models via reading comprehension](#). In *International Conference on Learning Representations*, volume 2024, pages 48624–48652.

DeepSeek-AI, Anyi Xu, Bangcai Lin, Bing Xue, Bingxuan Wang, Bingzheng Xu, Bochao Wu, Bowei Zhang, Chaofan Lin, Chen Dong, Chenchen Ling, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chengyu Hou, Chenhao Xu, Chenze Shao, Chong Ruan, Conner Sun, and 300 others. 2026. [Deepseek-v4: Towards highly efficient million-token context intelligence](#). *Preprint*, arXiv:2606.19348.

Stefan Depeweg, Jose-Miguel Hernandez-Lobato, Finale Doshi-Velez, and Steffen Udluft. 2018. [Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning](#). In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1184–1193. PMLR.

Andrey Goncharov, Daniil Vyazhev, Petr Sychev, Edvard Khalafyan, and Alexey Zaytsev. 2026. [Complexity-aware fine-tuning](#). *Preprint*, arXiv:2506.21220.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, and 542 others. 2024. [The llama 3 herd of models](#). *Preprint*, arXiv:2407.21783.

Yuxian Gu, Li Dong, Furu Wei, and Minlie Huang. 2024. [Minillm: Knowledge distillation of large language models](#). In *International Conference on Learning Representations*, volume 2024, pages 32694–32717.

Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. [Distilling the knowledge in a neural network](#). *Preprint*, arXiv:1503.02531.

Cheng-Yu Hsieh, Chun-Liang Li, Chih-kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alex Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. 2023. [Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes](#). In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 8003–8017, Toronto, Canada. Association for Computational Linguistics.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. [Lora: Low-rank adaptation of large language models](#). *Preprint*, arXiv:2106.09685.

Yaning Jia, Chunhui Zhang, Xingjian Diao, Xiangchi Yuan, Zhongyu Ouyang, Chiyu Ma, and Soroush Vosoughi. 2026. [What makes a good curriculum? disentangling the effects of data ordering on LLM mathematical reasoning](#). In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 34472–34488, San Diego, California, United States. Association for Computational Linguistics.

Wangyi Jiang, Yaojie Lu, Hongyu Lin, Xianpei Han, and Le Sun. 2025. [Teach small models to reason by curriculum distillation](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 7412–7422, Suzhou, China. Association for Computational Linguistics.

Xiaoqi Jiao, Yichun Yin, Lifeng Shang, Xin Jiang, Xiao Chen, Linlin Li, Fang Wang, and Qun Liu. 2020.

TinyBERT: Distilling BERT for natural language understanding. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 4163–4174, Online. Association for Computational Linguistics.

Hoang Anh Just, Myeongseob Ko, and Ruoxi Jia. 2026. The signal is in the steps: Local scoring for reasoning data selection. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 55125–55155. PMLR.

Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. 2020. Scaling laws for neural language models. *Preprint*, arXiv:2001.08361.

Jongwoo Ko, Sungnyun Kim, Tianyi Chen, and Se-Young Yun. 2024. DistiLLM: Towards streamlined distillation for large language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 24872–24895. PMLR.

Jierui Li and Raymond Mooney. 2024. Distilling algorithmic reasoning from LLMs via explaining solution programs. *Preprint*, arXiv:2404.08148.

Liunian Harold Li, Jack Hessel, Youngjae Yu, Xiang Ren, Kai-Wei Chang, and Yejin Choi. 2023. Symbolic chain-of-thought distillation: Small models can also “think” step-by-step. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2665–2679, Toronto, Canada. Association for Computational Linguistics.

Yang Li, Youssef Emad, Karthik Padthe, Jack Lanchantin, Weizhe Yuan, Thao Nguyen, Jason Weston, Shang-Wen Li, Dong Wang, Ilia Kulikov, and Xian Li. 2025. Naturalthoughts: Selecting and distilling reasoning traces for general reasoning tasks. *Preprint*, arXiv:2507.01921.

Wei Lu, Rachel K. Luu, and Markus J. Buehler. 2024. Fine-tuning large language models for domain adaptation: Exploration of training strategies, scaling, model merging and synergistic capabilities. *Preprint*, arXiv:2409.03444.

Andrey Malinin and Mark Gales. 2021. Uncertainty estimation in autoregressive structured prediction. *Preprint*, arXiv:2002.07650.

Microsoft, :, Abdelrahman Abouelenin, Atabak Ashfaq, Adam Atkinson, Hany Awadalla, Nguyen Bach, Jianmin Bao, Alon Benhaim, Martin Cai, Vishrav Chaudhary, Congcong Chen, Dong Chen, Dongdong Chen, Junkun Chen, Weizhu Chen, Yen-Chun Chen, Yi ling Chen, Qi Dai, and 57 others. 2025. Phi-4-mini technical report: Compact yet powerful multimodal language models via mixture-of-loras. *Preprint*, arXiv:2503.01743.

Patrik Okanovic, Roger Waleffe, Vasilis Mageirakos, Konstantinos E. Nikolakakis, Amin Karbasi, Dionysis Kalogieras, Nezihe Merve Gürel, and Theodoros Rekatsinas. 2023. Repeated random sampling for minimizing the time-to-accuracy of learning. *Preprint*, arXiv:2305.18424.

Qwen-Team. 2024. Qwen2.5: A party of foundation models.

Bakbergen Ryskulov, Iker García-Ferrero, David Montero, David Jansen, Ali Hashemi, Jezabel R. Garcia, Antonio Tiene, and Román Orús. 2026. Efficient knowledge distillation for llms: Offline top-k logits and a fused chunked kl loss. *Preprint*, arXiv:2608.03796.

Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2020. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. *Preprint*, arXiv:1910.01108.

Zhongwei Wan, Xin Wang, Che Liu, Samiul Alam, Yu Zheng, Jiachen Liu, Zhongnan Qu, Shen Yan, Yi Zhu, Quanlu Zhang, Mosharaf Chowdhury, and Mi Zhang. 2024. Efficient large language models: A survey. *Preprint*, arXiv:2312.03863.

Haozhe Wang, Haoran Que, Qixin Xu, Minghao Liu, Wangchunshu Zhou, Jiazhan Feng, Wanjun Zhong, Wei Ye, Tong Yang, Wenhao Huang, Ge Zhang, and Fangzhen Lin. 2025a. Reverse-engineered reasoning for open-ended generation. *Preprint*, arXiv:2509.06160.

Tianyang Wang, Yunze Wang, Jun Zhou, Benji Peng, Xinyuan Song, Charles Zhang, Xintian Sun, Qian Niu, Junyu Liu, Silin Chen, Keyu Chen, Ming Li, Pohsun Feng, Ziqian Bi, Ming Liu, Yichao Zhang, Cheng Fei, Caitlyn Heqi Yin, and Lawrence KQ Yan. 2025b. From aleatoric to epistemic: Exploring uncertainty quantification techniques in artificial intelligence. *Preprint*, arXiv:2501.03282.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Preprint*, arXiv:2406.01574.

Shuyao Xu, Cheng Peng, Jiangxuan Long, Weidi Xu, Wei Chu, and Yuan Qi. 2026. Harnessing negative signals: Reinforcement distillation from teacher data for LLM reasoning. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1618–1639, San Diego, California, United States. Association for Computational Linguistics.

Xiaohan Xu, Ming Li, Chongyang Tao, Tao Shen, Reynold Cheng, Jinyang Li, Can Xu, Dacheng Tao, and Tianyi Zhou. 2024. A survey on knowledge distillation of large language models. *Preprint*, arXiv:2402.13116.

Cehao Yang, Xueyuan Lin, Xiaojun Wu, Chengjin Xu, Xuhui Jiang, Honghao Liu, Hui Xiong, and Jian Guo. 2026. [Select2Reason: Efficient instruction-tuning data selection for long-CoT reasoning](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 6656–6671, San Diego, California, United States. Association for Computational Linguistics.

Ling Yang, Zhaochen Yu, Tianjun Zhang, Minkai Xu, Joseph E Gonzalez, Bin CUI, and Shuicheng YAN. 2025. [Supercorrect: Advancing small llm reasoning with thought template distillation and self-correction](#). In *International Conference on Learning Representations*, volume 2025, pages 3386–3418.

Mengxiang Zhang and Lingyuan Liu. 2026. [Calibrated progressive distillation: Co-designing curriculum and target mixing for knowledge distillation of large language models](#). In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 6757–6776, San Diego, California, United States. Association for Computational Linguistics.

Yifei Zhang, Bo Pan, Chen Ling, Yuntong Hu, and Liang Zhao. 2024. [ELAD: Explanation-guided large language models active distillation](#). In *Findings of the Association for Computational Linguistics: ACL 2024*, pages 4463–4475, Bangkok, Thailand. Association for Computational Linguistics.

Qinhong Zhou, Zonghan Yang, Peng Li, and Yang Liu. 2023. [Bridging the gap between decision and logits in decision-based knowledge distillation for pre-trained language models](#). In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13234–13248, Toronto, Canada. Association for Computational Linguistics.

A Experimental Hyperparameters

Hyperparameter	Value
Optimizer	AdamW (adamw_torch)
Learning rate	10^{-4}
Learning-rate schedule	Cosine
Warmup ratio	0.06
Weight decay	0
Adam (β_1, β_2)	(0.9, 0.999)
Adam epsilon	10^{-8}
Maximum gradient norm	1.0
Mixed precision	BF16 enabled
Gradient checkpointing	Enabled
torch.compile	Enabled
LoRA rank r	8
LoRA alpha	16
LoRA scaling α/r	2
LoRA dropout	0.1
Initialization	Gaussian

B Single-token entropy

Single-token entropy (Figure 4) measures a model’s uncertainty over the first answer token

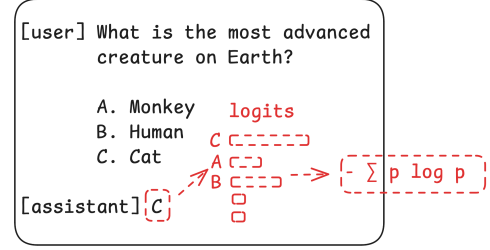


Figure 4: Single-token entropy example

when prompted to answer a question with a single token. For a question x , we apply softmax to the first-token logits to obtain probabilities $p(v | x)$ over the model’s full vocabulary V , and compute normalized Shannon entropy:

$$H(x) = -\frac{1}{\log |V|} \sum_{v \in V} p(v | x) \log p(v | x). \quad (2)$$

Dividing by $\log |V|$ puts student and proxy entropies on a common $[0, 1]$ scale despite differences in vocabulary size: values near zero indicate a concentrated prediction, while values near one indicate a nearly uniform distribution.

C Synthetic trace generation methods

We compare five ways to construct reasoning-and-answer targets for the same multiple-choice questions. Directly generated option letters are checked against ground-truth labels.

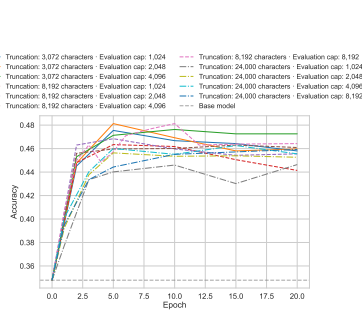
Direct reasoning trace. The teacher receives the question and options, generates reasoning, and predicts an option without seeing the ground-truth answer. The baseline retains both correct and incorrect responses.

Direct reasoning trace (correct only). We retain only baseline responses whose predicted option matches the ground truth, reducing the training set by discarding failed attempts.

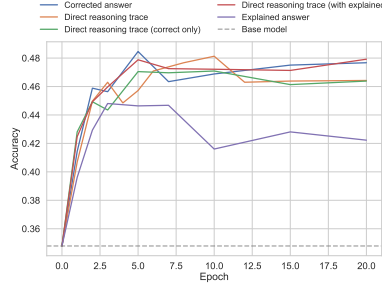
Explained answer. The teacher receives the question, options, and correct answer. It generates a step-by-step explanation that reads as a fresh solution, allowing alternatives and self-correction while avoiding acknowledgment of the supplied answer. The target pairs this explanation with the ground-truth option.

Direct reasoning trace (with explained). We retain correct baseline responses and replace each

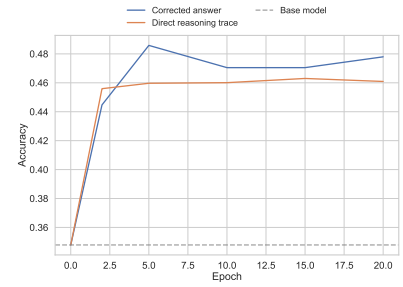
553	incorrect response with the explained-answer re-		
554	sponse for the same question. This preserves ques-		
555	tion coverage while restricting answer-conditioned		
556	generation to failed attempts.		
557	Corrected answer. For each incorrect baseline		
558	response, the teacher receives the original reason-		
559	ing, its incorrect option, and the correct option. It		
560	continues the trace toward the correct answer, pre-		
561	sending a natural self-correction without restarting		
562	or repeating the original text. We append the con-		
563	tinuation and use the ground-truth option as the		
564	final answer. Correct baseline responses remain		
565	intact.		
566	C.1 Prompt templates		
567	The placeholders below are filled with the subject,		
568	question, and options. Correct-only filtering and		
569	replacement with explanations reuse these prompts;		
570	they introduce no additional generation prompt.		
571	Direct generation: system prompt. Reasoning		
572	is collected separately from the teacher’s final an-		
573	swer.		
574	The following are multiple choice questions about		
575	{subject}. Choose a correct option letter. Answer		
576	with a single symbol. Do not print anything else.		
577	Explained answer: system prompt.		
578	The following is a multiple choice question about		
579	{subject}. The correct answer is provided to you,		
580	but your task is to produce a detailed chain-of-		
581	thought with step-by-step reasoning that reads as		
582	if you were solving the question from scratch—		
583	including any natural uncertainty, consideration		
584	of multiple options, or self-correction. Do not		
585	state or imply that the answer was given to you.		
586	End your response with Answer: <letter>.		
587	Corrected answer: system prompt.		
588	The following are multiple choice questions about		
589	{subject}. You will be shown a question, a partial		
590	chain-of-thought that was previously produced		
591	for it, the incorrect option that chain-of-thought		
592	arrived at, and the correct option letter. Continue		
593	the partial chain-of-thought from where it left off		
594	so that it naturally arrives at the correct option.		
595	Do not restart the reasoning, do not repeat what		
596	was already written, and do not acknowledge that		
597	the previous attempt was wrong or that the correct		
598	answer was given to you. Write the continua-		
599	tion as if you were the same reasoner noticing a		
600	mistake or new consideration mid-thought and re-		
601	vising course. In the end, answer with the correct		
602	option letter. End your response with Answer:		
603	<letter>.		
	User prompts. All three begin with:		
	Question: {question}		
	Options:		
	a. {option text}		
	b. {option text}		
	...		
	For explained answers, append Correct option:		
	followed by the ground-truth letter. For corrections,		
	append:		
	Partial reasoning so far:		
	{original reasoning}		
	Incorrect option this reasoning led to:		
	{predicted letter}		
	Correct option: {ground-truth letter}		
	C.2 Illustrative examples		
	For brevity, consider a toy question with two op-		
	tions: “A rectangle has sides of length 3 and 2.		
	What is its perimeter? a. 6; b. 10.” The following		
	illustrative targets show how the methods handle		
	an incorrect attempt.		
	Direct trace. “The perimeter is $3 \times 2 = 6$. An-		
	swer: a.” The baseline retains this response; the		
	correct-only variant discards it.		
	Explained answer. Given correct option b, a		
	fresh explanation is: “There are two sides of each		
	length, so the perimeter is $2(3 + 2) = 10$. An-		
	swer: b.” The with-explained variant substitutes		
	this response for the incorrect direct trace.		
	Corrected answer. Given the original reason-		
	ing, incorrect option a, and correct option b, a		
	continuation is: “That product gives the area.		
	For the perimeter, I need to add all four sides:		
	$3 + 2 + 3 + 2 = 10$. Answer: b.” The target retains		
	the original reasoning, appends this continuation,		
	and ends with b.		
	C.3 Per-model results		
	Figures 5–7 show test accuracy across training		
	epochs for each student. The left panels compare		
	head-truncation limits of 3,072, 8,192, and 24,000		
	characters for direct reasoning traces; each legend		
	separately specifies the evaluation cap in tokens.		
	The generation-method panels hold head truncation		
	fixed at 8,192 characters and compare methods at		
	the stated evaluation cap. Dashed horizontal lines		
	and epoch-zero values use the base model’s ac-		
	curacy prompted to answer with chain-of-thought		
	(base models originally are not reasoners), shared		
	across evaluation budgets.		



(a) Head-truncation comparison

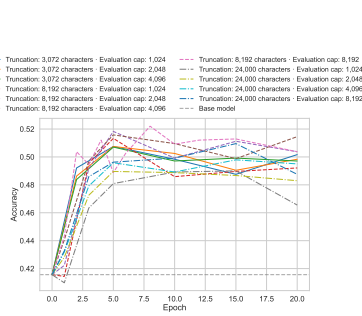


(b) Generation methods: truncation 8,192, eval cap 8,192

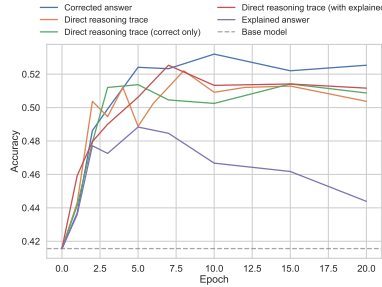


(c) Generation methods: truncation 8,192, eval cap 4,096

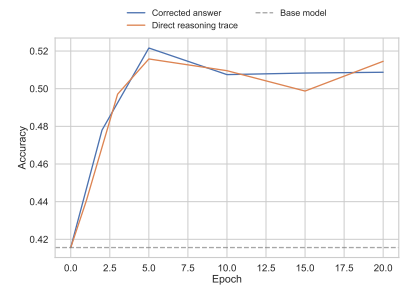
Figure 5: Synthetic-trace distillation results for LLaMA 3.2 3B.



(a) Head-truncation comparison



(b) Generation methods: truncation 8,192, eval cap 8,192



(c) Generation methods: truncation 8,192, eval cap 4,096

Figure 6: Synthetic-trace distillation results for Qwen 2.5 3B.

Budget (chars)	Direct traces	Corrected traces
3,072	3,603 (37.4%)	3,721 (38.7%)
8,192	1,879 (19.5%)	1,921 (20.0%)
24,000	738 (7.7%)	749 (7.8%)

Table 3: Number (percentage) of training traces exceeding each truncation budget

D Synthetic trace truncation and reasoning caps

Training truncation limits the retained teacher reasoning in characters; the reasoning cap limits the student’s generated reasoning at evaluation time in tokens. These are separate controls: a 3,072-character training trace can be evaluated with a 4,096-token cap. The final answer letter is retained separately when training traces are shortened.

Table 3 reports how many training traces exceed each character budget before truncation.

D.1 Which part of the trace is retained?

Head truncation keeps a prefix. Middle truncation keeps an initial segment and a final segment, dropping the interior of an overlong trace. Tail truncation keeps only a suffix. At the nominal 24,000-character setting, head retains up to 24,000

characters, middle retains the first 18,000 and last 6,000 of overlong traces, and tail retains the last 24,000. For corrected traces, middle truncation instead reserves up to 1,600 characters for the correction and uses the remaining space for the original prefix.

Figure 8 compares these policies at a fixed 8,192-token evaluation cap. Preferences vary by model: head gives the best observed checkpoint for LLaMA and Qwen, while tail does so for Phi4-mini.

D.2 Training truncation length versus evaluation cap

For direct traces with head truncation at 3,072, 8,192, or 24,000 characters, we compare available evaluation caps of 1,024, 2,048, 4,096, and 8,192 tokens. Figures 9 and 10 separate training length from evaluation budget. Best-checkpoint accuracy is the maximum over observed trained checkpoints for each cap; last-checkpoint accuracy uses the final available epoch. Only evaluated combinations are plotted.

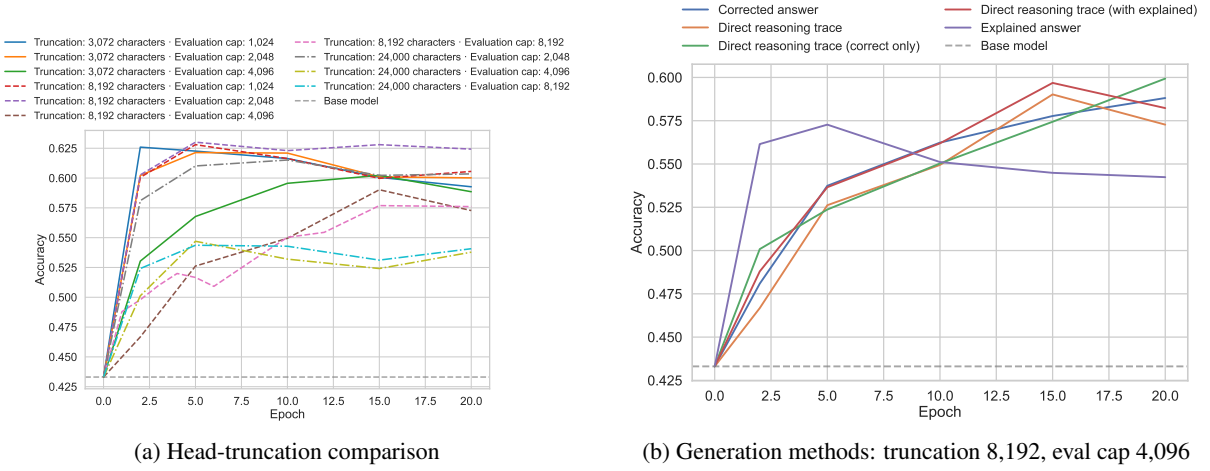


Figure 7: Synthetic-trace distillation results for Phi4-mini.

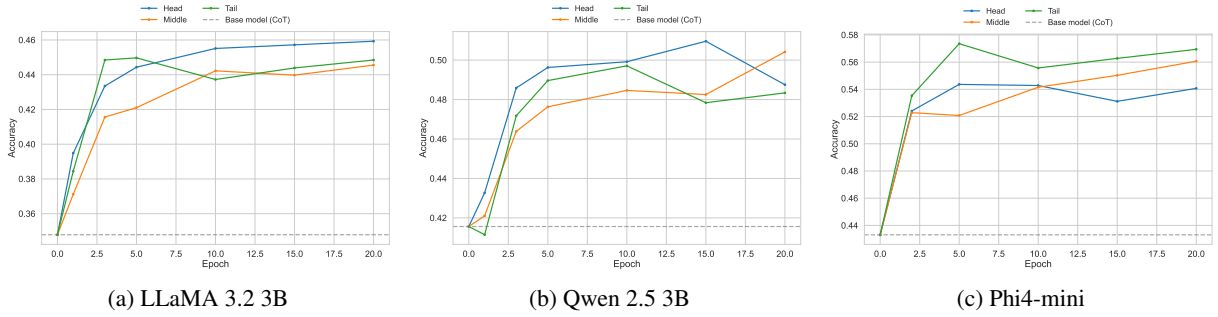


Figure 8: Training truncation policies at the nominal 24,000-character setting, evaluated with an 8,192-token reasoning cap. Dashed lines and epoch-zero points give the base model’s chain-of-thought accuracy.

E Sensitivity Study Details

do not increase it.

The sampler ablation holds the corrected traces and the per-epoch mix of 1,024 reasoning examples and 256 answer-only examples fixed, varying the selection rule: uniform random sampling or sampling without replacement weighted by student entropy, positive entropy gain, or discounted entropy gain. The discounted score uses $k(t) = 0.8 \cdot 0.01^{t/100}$, where t is the resampling epoch; its proxy contribution falls to 0.008 at epoch 100. It is labelled “Combined entropy (sampled)” in the graphs. Table 4 reports final accuracy; Figure 14 compares the weighted samplers and random sampling.

Figures 11–13 provide individual accuracy and coverage curves. Lines average available seed replicas. Horizontal axes count reasoning-question selections, excluding the additional answer-only examples: 1,024 per standard epoch, k for a fixed- k run, and 9,626 for full-dataset baselines. Accuracy plots share each model’s initial chain-of-thought score across evaluation budgets.

Coverage counts distinct question IDs selected at least once, so repeated epochs and response formats

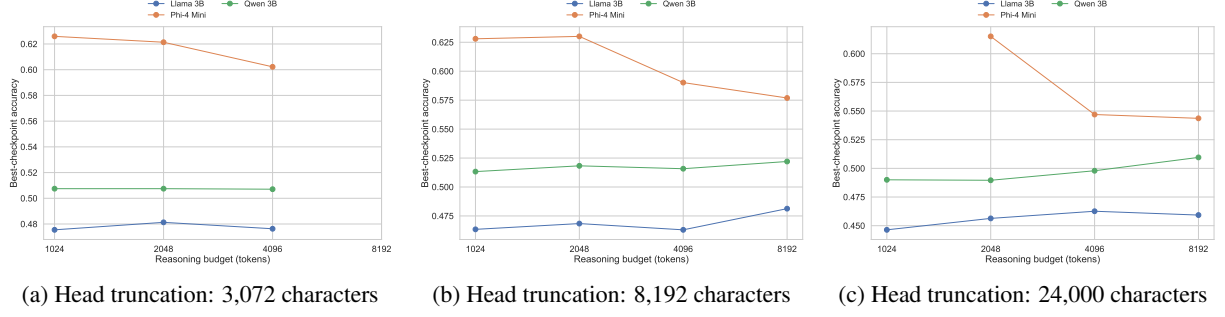


Figure 9: Best-checkpoint accuracy versus evaluation reasoning cap for each training truncation length.

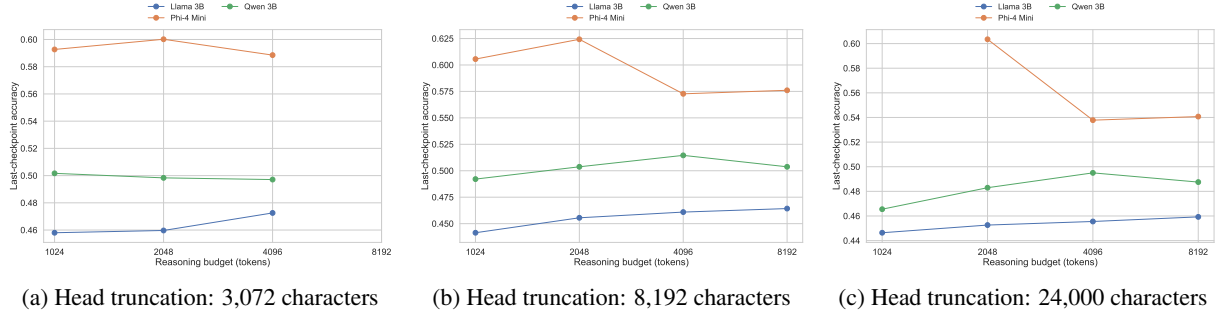


Figure 10: Last-checkpoint accuracy versus evaluation reasoning cap for each training truncation length.

Sampler	LLaMA 3.2 3B	Qwen 2.5 3B	Phi4-mini	Mean
Direct reasoning trace (baseline)	46.09	51.45	57.27	51.61
Random	51.77 $\pm$ 2.76	57.91 $\pm$ 1.40	55.57 $\pm$ 2.54	55.08 $\pm$ 0.80
Student entropy	51.18 $\pm$ 3.11	57.32 $\pm$ 1.52	59.12 $\pm$ 5.22	55.87 $\pm$ 1.40
Discounted entropy gain	51.50	56.90	<u>59.53</u> $\pm$ 2.11	<u>55.98</u>
Entropy gain	52.58 $\pm$ 1.94	56.64 $\pm$ 3.79	62.22 $\pm$ 2.42	57.15 $\pm$ 1.01

Table 4: Distillation pipeline last epoch accuracy for different sampling methods (4096-token thinking budget)

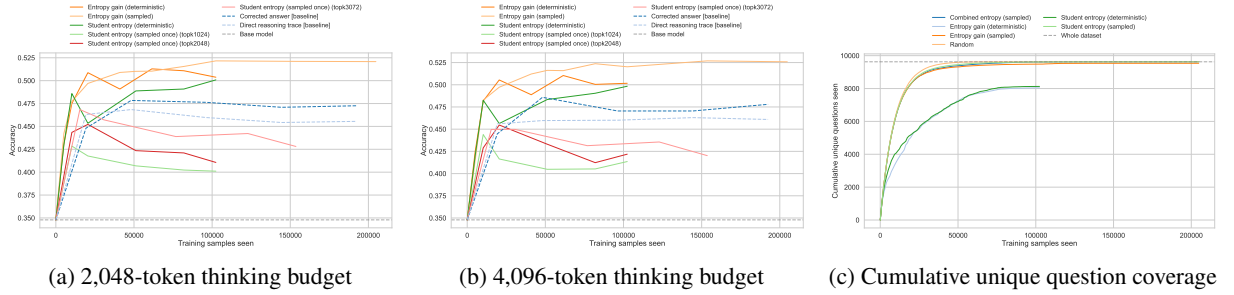


Figure 11: Distillation by sampling strategy for LLaMA 3.2 3B.

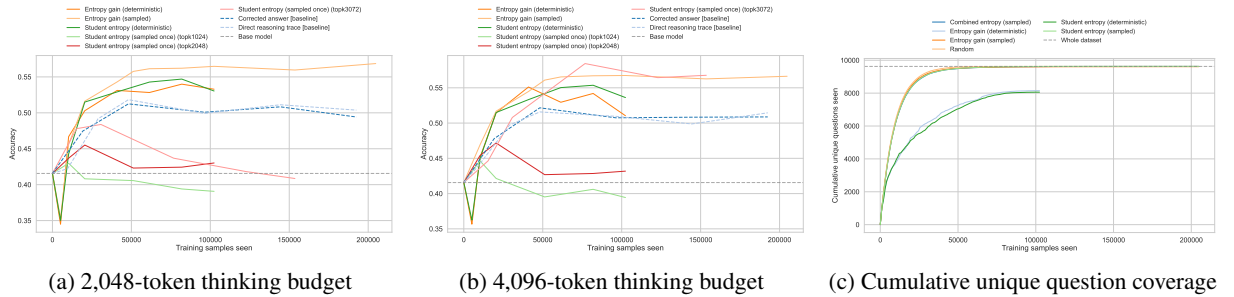


Figure 12: Distillation by sampling strategy for Qwen 2.5 3B.

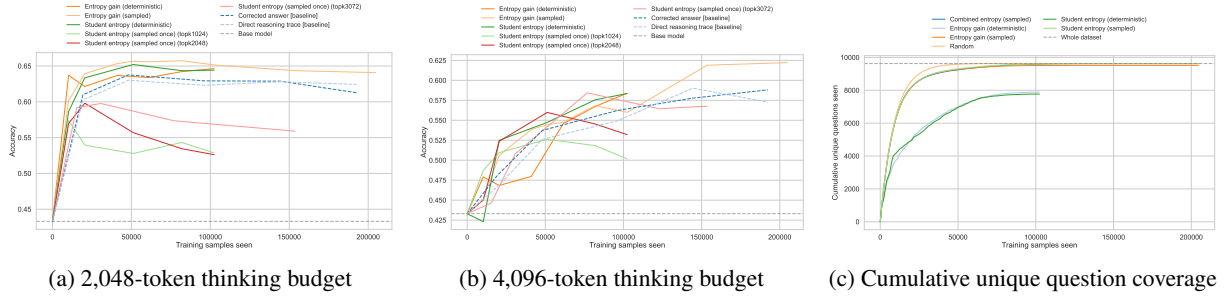


Figure 13: Distillation by sampling strategy for Phi4-mini.

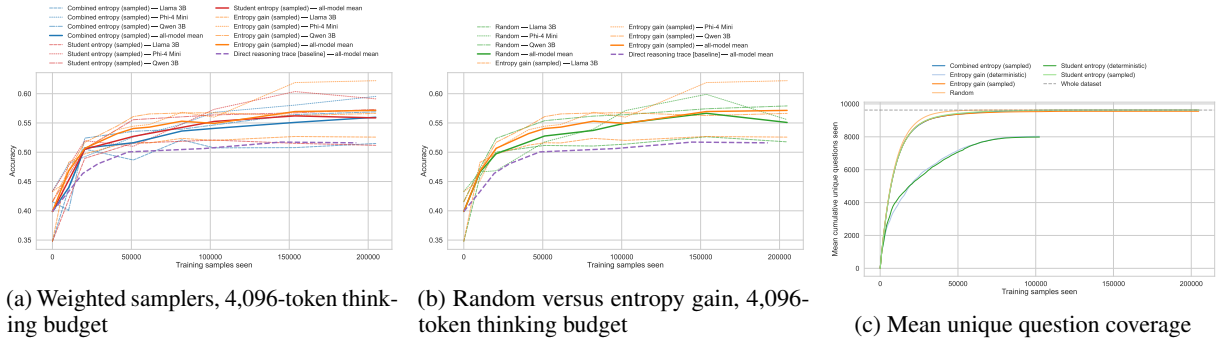


Figure 14: Sampler comparisons and question coverage across all three models. In the accuracy panels, thin dashed curves show individual models, thick solid curves their equal-weight means, and the thick dashed curve the direct-trace baseline mean.